

Nand2Tetris - Project 1

Boolean Logic Gates - HDL Code (code + simulator screenshots)

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1. Not.hdl

```
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/Not.hdl
/**
 * Not gate:
 * if (in) out = 0, else out = 1
 */
CHIP Not {
    IN in;
    OUT out;

    PARTS:
        Nand(a = in, b=in ,out=out);
        //// Replace this comment with your code.
}
```

Simulator screenshot

The screenshot shows the IAND2Tetris Hardware Simulator interface. The top bar displays 'IAND2Tetris / Hardware Simulator / Not.hdl'. Below this, the HDL code for the Not gate is shown in a text editor. The code is as follows:

```
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/1/Not.hdl
5 /**
6  * Not gate:
7  * if (in) out = 0, else out = 1
8  */
9 CHIP Not {
10     IN in;
11     OUT out;
12
13     PARTS:
14         Nand(a = in, b=in ,out=out);
15         //// Replace this comment with your code.
16 }
```

On the right side of the simulator, the 'Chip Not' is shown with its input and output pins. The input pin 'in' is set to 1, and the output pin 'out' is set to 0. Below the chip, the 'Test' section shows the current state of the simulation: 'Steps: 1 Outputs: 3'. The 'Test Script' section shows the following code:

```
9
10 set in 0,
11 eval,
12 output;
13
14 set in 1,
15 eval,
```

2. And.hdl

```
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/And.hdl
/**
 * And gate:
 * if (a and b) out = 1, else out = 0
 */
CHIP And {
    IN a, b;
    OUT out;

    PARTS:
        Nand(a=a,b=b,out=nandout);
        Not(in=nandout, out=out);
        //// Replace this comment with your code.
}
```

Simulator screenshot

The screenshot shows the VAND2Tetris Hardware Simulator interface. The top bar displays the project name "Project 1" and the selected file "And.hdl". The main window is divided into two panes. The left pane shows the HDL code for the "And" chip, which is a simple AND gate implementation using a NAND gate and a NOT gate. The right pane shows the chip's internal state, including input pins (a=1, b=0), output pins (out=0), and internal pins (nandout=1). The bottom status bar indicates the simulation is running at "Steps: 1" with "Outputs: 0". The "Test" tab is active, showing a test script that sets input b to 0, evaluates the chip, and outputs the result.

VAND2Tetris / Hardware Simulator / And.hdl

HDL ☐ --LBSt **Project 1** **And** **Chip And** Eval Reset Clock: 0

```
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/1/And.hdl
5 /**
6  * And gate:
7  * if (a and b) out = 1, else out = 0
8  */
9 CHIP And {
10     IN a, b;
11     OUT out;
12
13     PARTS:
14         Nand(a=a,b=b,out=nandout);
15         Not(in=nandout, out=out);
16 }
```

Input pins
a 1
b 0

Output pins
out 0

Internal pins
nandout 1

Test **Steps: 1** **Outputs: 0** Slow Fast

Test Script Compare File Output File Diff Table

```
21 set b 0,
22 eval,
23 output;
24
25 set a 1,
26 set b 1,
27 eval,
```

3. Or.hdl

```
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/Or.hdl
/**
 * Or gate:
 * if (a or b) out = 1, else out = 0
 */
CHIP Or {
    IN a, b;
    OUT out;

    PARTS:
        Not(in=a,out=nota);
        Not(in=b,out=notb);
        Nand(a=nota,b=notb, out=out);
    //// Replace this comment with your code.
}
```

Simulator screenshot

The screenshot shows the IAND2Tetris Hardware Simulator interface. The top bar displays the project name "Project 1" and the selected file "Or.hdl". The main editor area shows the HDL code for the Or gate, which is currently commented out. The right panel shows the simulation results for the "Chip Or" component. The input pins are "a" (1) and "b" (0). The output pin is "out" (1). The internal pins are "nota" (0) and "notb" (1). The bottom panel shows the test script, which is currently empty.

Test Script

```
21 set b 0,
22 eval,
23 output;
24
25 set a 1,
26 set b 1,
27 eval.
```

4. Xor.hdl

```
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/Xor.hdl
/**
 * Exclusive-or gate:
 * if ((a and Not(b)) or (Not(a) and b)) out = 1, else out = 0
 */
CHIP Xor {
    IN a, b;
    OUT out;

    PARTS:
    //// Replace this comment with your code.
    Not(in=a,out=nota);
    Not(in=b,out=notb);
    And(a=a,b=notb,out=aAndNotb);
    And(a=nota,b=b,out=notaAndb);
    Or(a=aAndNotb,b=notaAndb,out=out);
}
```

Simulator screenshot

The screenshot shows the Nand2Tetris Hardware Simulator interface. The top bar displays "NAND2Tetris / Hardware Simulator / Xor.hdl". The main window is divided into three sections:

- Code Editor:** Displays the HDL code for the Xor chip. The code is as follows:

```
5 /**
7 * if ((a and Not(b)) or (Not(a) and b)) out = 1, else out = 0
8 */
9 CHIP Xor {
10     IN a, b;
11     OUT out;
12
13     PARTS:
14     //// Replace this comment with your code.
15     Not(in=a,out=nota);
16     Not(in=b,out=notb);
17     And(a=a,b=notb,out=aAndNotb);
18     And(a=nota,b=b,out=notaAndb);
19     Or(a=aAndNotb,b=notaAndb,out=out);
20 }
```
- Input pins:** Shows the current values of the input pins: `a` is 1, `b` is 0.
- Output pins:** Shows the current value of the output pin: `out` is 1.
- Internal pins:** Shows the current values of the internal pins: `nota` is 0, `notb` is 1, `aAndNotb` is 1, and `notaAndb` is 0.

At the bottom, the **Test** tab is active, showing a test script with the following steps:

```
21 set b 0,
22 eval,
23 output;
24
25 set a 1,
26 set b 1,
27 eval,
```

The test script is currently at step 1, and the output is 0. The speed is set to Slow.

5. Mux.hdl

```
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/Mux.hdl
/**
 * Multiplexor:
 * if (sel = 0) out = a, else out = b
 */
CHIP Mux {
    IN a, b, sel;
    OUT out;

    PARTS:
    //// Replace this comment with your code.
    Not(in=sel,out=nSel);
    And(a=a,b=nSel,out=x);
    And(a=b, b=sel, out=y);
    Or(a=x,b=y,out=out);
}
```

Simulator screenshot

The screenshot shows the Nand2Tetris Hardware Simulator interface. The top bar displays "NAND2Tetris / Hardware Simulator / Mux.hdl". The main window is divided into two panes. The left pane shows the HDL code for the Mux chip, with line numbers 5 through 19. The code implements a multiplexer that selects between inputs 'a' and 'b' based on the selector 'sel'. The right pane shows the simulation results for the "Chip Mux". It lists input pins: 'a' (1), 'b' (1), and 'sel' (0). Output pins are 'out' (1). Internal pins are 'nSel' (1), 'x' (1), and 'y' (1). The bottom status bar indicates "Test" mode, "Steps: 1", "Outputs: 0", and a speed slider between "Slow" and "Fast". The bottom-most pane shows a test script with the following commands:

```
21 set b 1,
22 set sel 0,
23 eval,
24 output;
25
26 set sel 1,
27 eval,
```

6. DMux.hdl

```
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/DMux.hdl
/**
 * Demultiplexor:
 * [a, b] = [in, 0] if sel = 0
 *          [0, in] if sel = 1
 */
CHIP DMux {
    IN in, sel;
    OUT a, b;

    PARTS:
    //// Replace this comment with your code.
    Not(in =sel,out = nsel);
    And(a = in,b = nsel,out=a);
    And(a = in, b=sel ,out=b);
}
```

Simulator screenshot

The screenshot shows the Nand2Tetris Hardware Simulator interface. The top bar displays "NAND2Tetris / Hardware Simulator / DMux.hdl". The main window is divided into several sections:

- HDL Editor:** Contains the Verilog code for the DMux chip, including comments and the main logic.
- Chip DMux:** A panel on the right showing the chip's configuration. It includes input pins (in, sel), output pins (a, b), and internal pins (nsel). The values for these pins are displayed in blue boxes.
- Test Panel:** Located at the bottom, it shows the test script and the simulation results. The test script includes lines for setting inputs and evaluating outputs.

The simulation results show that the chip is currently in state 0, with inputs in=1 and sel=1, and outputs a=0 and b=1. The internal pin nsel is also shown as 0.

7. Not16.hdl

```
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/01/Not16.hdl
/**
 * 16-bit Not gate:
 * for i = 0, ..., 15:
 * out[i] = Not(a[i])
 */
CHIP Not16 {
    IN in[16];
    OUT out[16];

    PARTS:
    //// Replace this comment with your code.
    Not(in = in[0],out = out[0]);
    Not(in = in[1],out = out[1]);
    Not(in = in[2],out = out[2]);
    Not(in = in[3],out = out[3]);
    Not(in = in[4],out = out[4]);
    Not(in = in[5],out = out[5]);
    Not(in = in[6],out = out[6]);
    Not(in = in[7],out = out[7]);
    Not(in = in[8],out = out[8]);
    Not(in = in[9],out = out[9]);
    Not(in = in[10],out = out[10]);
    Not(in = in[11],out = out[11]);
    Not(in = in[12],out = out[12]);
    Not(in = in[13],out = out[13]);
    Not(in = in[14],out = out[14]);
    Not(in = in[15],out = out[15]);
}
```

Simulator screenshot

The screenshot shows the Nand2Tetris Hardware Simulator interface. The top bar displays "NAND2Tetris / Hardware Simulator / Not16.hdl". Below this, the "HDL" tab is active, showing the code for the "Not16" chip. The code is a 16-bit NOT gate implementation. The "Chip Not16" panel on the right shows the input pins as "1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0" and the output pins as "0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1". The "Test" panel at the bottom shows the test script for the chip.

HDL --LBSt Project 1 Not16

```
18 CHIP Not16 {
19     Not(in = in[2],out = out[2]);
20     Not(in = in[3],out = out[3]);
21     Not(in = in[4],out = out[4]);
22     Not(in = in[5],out = out[5]);
23     Not(in = in[6],out = out[6]);
24     Not(in = in[7],out = out[7]);
25     Not(in = in[8],out = out[8]);
26     Not(in = in[9],out = out[9]);
27     Not(in = in[10],out = out[10]);
28     Not(in = in[11],out = out[11]);
29     Not(in = in[12],out = out[12]);
30     Not(in = in[13],out = out[13]);
31     Not(in = in[14],out = out[14]);
32     Not(in = in[15],out = out[15]);
33 }
```

Chip Not16 Eval Reset Clock: 0

Input pins
in 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 dec

Output pins
out 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec

Test Steps: 1 Outputs: 0 Slow Fast

Test Script Compare File Output File Diff Table

```
21
22 set in %B0011110011000011,
23 eval,
24 output;
25
26 set in %B0001001000110100,
27 eval,
```

8. And16.hdl

```
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/And16.hdl
/**
 * 16-bit And gate:
 * for i = 0, ..., 15:
 *   out[i] = a[i] And b[i]
 */
CHIP And16 {
    IN a[16], b[16];
    OUT out[16];

    PARTS:
    //// Replace this comment with your code.
    And(a=a[0],b=b[0],out=out[0]);
    And(a=a[1],b=b[1],out=out[1]);
    And(a=a[2],b=b[2],out=out[2]);
    And(a=a[3],b=b[3],out=out[3]);
    And(a=a[4],b=b[4],out=out[4]);
    And(a=a[5],b=b[5],out=out[5]);
    And(a=a[6],b=b[6],out=out[6]);
    And(a=a[7],b=b[7],out=out[7]);
    And(a=a[8],b=b[8],out=out[8]);
    And(a=a[9],b=b[9],out=out[9]);
    And(a=a[10],b=b[10],out=out[10]);
    And(a=a[11],b=b[11],out=out[11]);
    And(a=a[12],b=b[12],out=out[12]);
    And(a=a[13],b=b[13],out=out[13]);
    And(a=a[14],b=b[14],out=out[14]);
    And(a=a[15],b=b[15],out=out[15]);
}
```

Simulator screenshot

The screenshot shows the Nand2Tetris Hardware Simulator interface. The top bar displays "NAND2Tetris / Hardware Simulator / And16.hdl". The main window is divided into two panes. The left pane shows the HDL code for the "And16" chip, with lines 16 through 29 implementing 16-bit AND gates. The right pane shows the "Chip And16" configuration with input pins 'a' and 'b' set to 1010101010101010 and 0101010101010101 respectively, and the output pin 'out' showing 0000000000000000. The bottom status bar indicates "Steps: 1 Outputs: 0" and includes a speed slider. Below the main window, a "Test Script" pane shows the test code:

```
31 set b %00000111111110000,
32 eval,
33 output;
34
35 set a %0001001000110100,
36 set b %01001100001110110,
37 eval,
```

9. Or16.hdl

```
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/Or16.hdl
/**
 * 16-bit Or gate:
 * for i = 0, ..., 15:
 * out[i] = a[i] Or b[i]
 */
CHIP Or16 {
    IN a[16], b[16];
    OUT out[16];

    PARTS:
    //// Replace this comment with your code.
    Or(a=a[0],b=b[0],out=out[0]);
    Or(a=a[1],b=b[1],out=out[1]);
    Or(a=a[2],b=b[2],out=out[2]);
    Or(a=a[3],b=b[3],out=out[3]);
    Or(a=a[4],b=b[4],out=out[4]);
    Or(a=a[5],b=b[5],out=out[5]);
    Or(a=a[6],b=b[6],out=out[6]);
    Or(a=a[7],b=b[7],out=out[7]);
    Or(a=a[8],b=b[8],out=out[8]);
    Or(a=a[9],b=b[9],out=out[9]);
    Or(a=a[10],b=b[10],out=out[10]);
    Or(a=a[11],b=b[11],out=out[11]);
    Or(a=a[12],b=b[12],out=out[12]);
    Or(a=a[13],b=b[13],out=out[13]);
    Or(a=a[14],b=b[14],out=out[14]);
    Or(a=a[15],b=b[15],out=out[15]);
}
```

Simulator screenshot

The screenshot shows the Nand2Tetris Hardware Simulator interface. The top bar displays "NAND2Tetris / Hardware Simulator / Or16.hdl". The main window is divided into two panes. The left pane shows the HDL code for the Or16 chip, with lines 17 through 30 implementing the Or gate for each bit. The right pane shows the chip's configuration and test results. The "Chip Or16" is selected, and the "Eval" button has been pressed. The "Input pins" section shows two 16-bit inputs, 'a' and 'b', both set to 0. The "Output pins" section shows the 16-bit output 'out', which is 0. The "Test" section at the bottom shows the test script, which sets inputs 'a' and 'b' to specific bit patterns and evaluates the output.

Test Script

```
31 set b %B0000111111110000,
32 eval,
33 output;
34
35 set a %B0001001000110100,
36 set b %B1001100001110110,
37 eval,
```

10. Mux16.hdl

```
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/Mux16.hdl
/**
 * 16-bit multiplexor:
 * for i = 0, ..., 15:
 * if (sel = 0) out[i] = a[i], else out[i] = b[i]
 */
CHIP Mux16 {
    IN a[16], b[16], sel;
    OUT out[16];

    PARTS:
    //// Replace this comment with your code.
    Mux(a=a[0],b=b[0],sel=sel,out=out[0]);
    Mux(a=a[1],b=b[1],sel=sel,out=out[1]);
    Mux(a=a[2],b=b[2],sel=sel,out=out[2]);
    Mux(a=a[3],b=b[3],sel=sel,out=out[3]);
    Mux(a=a[4],b=b[4],sel=sel,out=out[4]);
    Mux(a=a[5],b=b[5],sel=sel,out=out[5]);
    Mux(a=a[6],b=b[6],sel=sel,out=out[6]);
    Mux(a=a[7],b=b[7],sel=sel,out=out[7]);
    Mux(a=a[8],b=b[8],sel=sel,out=out[8]);
    Mux(a=a[9],b=b[9],sel=sel,out=out[9]);
    Mux(a=a[10],b=b[10],sel=sel,out=out[10]);
    Mux(a=a[11],b=b[11],sel=sel,out=out[11]);
    Mux(a=a[12],b=b[12],sel=sel,out=out[12]);
    Mux(a=a[13],b=b[13],sel=sel,out=out[13]);
    Mux(a=a[14],b=b[14],sel=sel,out=out[14]);
    Mux(a=a[15],b=b[15],sel=sel,out=out[15]);
}
```

Simulator screenshot

The screenshot shows the Nand2Tetris Hardware Simulator interface. The top bar displays "NAND2Tetris / Hardware Simulator / Mux16.hdl". The main window is divided into two panes. The left pane shows the HDL code for the Mux16 chip, with lines 10 to 32 visible. The right pane shows the simulation results for the "Chip Mux16". It includes input pins 'a' and 'b' (both set to 1111111111111111), a select pin 'sel' (set to 1), and an output pin 'out' (set to 1111111111111111). The bottom status bar shows "Test Script" and "Compare File" buttons, along with "Steps: 1 Outputs: 0" and a speed slider. The bottom-most pane shows the test script code:

```
41 set b %00101010101010101,
42 set sel 0,
43 eval,
44 output;
45
46 set sel 1,
47 eval.
```

11. Or8Way.hdl

```
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/Or8Way.hdl
/**
 * 8-way Or gate:
 * out = in[0] Or in[1] Or ... Or in[7]
 */
CHIP Or8Way {
    IN in[8];
    OUT out;

    PARTS:
    ///// Replace this comment with your code.
    Or(a=in[0],b=in[1],out=o1);
    Or(a=in[2],b=in[3],out=o2);
    Or(a=in[4],b=in[5],out=o3);
    Or(a=in[6],b=in[7],out=o4);
    Or(a=o1,b=o2,out=o5);
    Or(a=o3,b=o4,out=o6);
    Or(a=o5,b=o6,out=out);
}
```

Simulator screenshot

The screenshot shows the IAND2Tetris Hardware Simulator interface. The top bar displays the project name "IAND2Tetris / Hardware Simulator / Or8Way.hdl". The main window is divided into three sections: a code editor on the left, a test script at the bottom, and a results panel on the right.

The code editor shows the HDL code for the Or8Way chip, with line numbers 5 through 22. The code defines an 8-input OR gate that outputs the OR of all inputs.

The test script at the bottom shows the following commands:

```
21
22 set in %B00000001,
23 eval,
24 output;
25
26 set in %B00100110,
27 eval,
```

The results panel on the right shows the current state of the chip. The input pins are set to 1 1 1 1 1 1 1 1. The output pins show the result of the OR operation, which is 1. The internal pins show the intermediate results of the OR operations.

12. Mux4Way16.hdl

```
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/Mux4Way16.hdl
/**
 * 4-way 16-bit multiplexor:
 * out = a if sel = 00
 *      b if sel = 01
 *      c if sel = 10
 *      d if sel = 11
 */
CHIP Mux4Way16 {
    IN a[16], b[16], c[16], d[16], sel[2];
    OUT out[16];

    PARTS:
    ///// Replace this comment with your code.
    Mux16(a=a, b=b, sel=sel[0],out=ab);
    Mux16(a=c, b=d, sel=sel[0],out=cd);
    Mux16(a=ab,b=cd,sel=sel[1], out=out);
}
```

Simulator screenshot

The screenshot displays the Nand2Tetris Hardware Simulator interface. The top bar shows the project name "Mux4Way16.hdl". The main window is divided into two panes. The left pane shows the HDL code for the Mux4Way16 chip, which is a 4-way 16-bit multiplexor. The code defines inputs a, b, c, d (16 bits each) and a 2-bit selector sel. It uses two Mux16 chips to implement the logic: the first Mux16 selects between a and b based on sel[0] to produce intermediate output ab, and the second Mux16 selects between c and d based on sel[0] to produce intermediate output cd. Finally, a third Mux16 selects between ab and cd based on sel[1] to produce the final 16-bit output out. The right pane shows the simulation results for the input pins (a, b, c, d, sel) and the output pin (out). All input pins are currently set to 1, and the output pin is also set to 1. The bottom status bar indicates "Steps: 1 Outputs: 0" and provides controls for simulation speed (Slow to Fast) and a test script.

```
5 /**
8 *      b if sel = 01
9 *      c if sel = 10
10 *      d if sel = 11
11 */
12 CHIP Mux4Way16 {
13     IN a[16], b[16], c[16], d[16], sel[2];
14     OUT out[16];
15
16     PARTS:
17     ///// Replace this comment with your code.
18     Mux16(a=a, b=b, sel=sel[0],out=ab);
19     Mux16(a=c, b=d, sel=sel[0],out=cd);
20     Mux16(a=ab,b=cd,sel=sel[1], out=out);
21 }
```

Input pins

Pin	Value	Label
a	1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1	dec
b	1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1	dec
c	1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1	dec
d	1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1	dec
sel	0 1	dec

Output pins

Pin	Value	Label
out	1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1	dec

Internal pins

Pin	Value	Label
ab	1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1	dec

Test Script Compare File Output File Diff Table

```
41
42 set sel 2,
43 eval,
44 output;
45
46 set sel 3,
47 eval,
```

13. Mux8Way16.hdl

```
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/Mux8Way16.hdl
/**
 * 8-way 16-bit multiplexor:
 * out = a if sel = 000
 *      b if sel = 001
 *      c if sel = 010
 *      d if sel = 011
 *      e if sel = 100
 *      f if sel = 101
 *      g if sel = 110
 *      h if sel = 111
 */
CHIP Mux8Way16 {
    IN a[16], b[16], c[16], d[16],
       e[16], f[16], g[16], h[16],
       sel[3];
    OUT out[16];

    PARTS:
    //// Replace this comment with your code.
    Mux4Way16(a=a, b=b, c=c, d=d, sel=sel[0..1], out=x);
    Mux4Way16(a=e, b=f, c=g, d=h, sel=sel[0..1], out=y);
    Mux16(a=x, b=y, sel=sel[2], out=out);
}
```

Simulator screenshot

The screenshot shows the Nand2Tetris Hardware Simulator interface. The top bar displays the project name "Mux8Way16.hdl". The main window is divided into two panes. The left pane shows the HDL code for the Mux8Way16 chip, which is a 8-way 16-bit multiplexor. The code defines inputs a, b, c, d, e, f, g, h (each 16 bits) and a 3-bit selector sel. It uses two Mux4Way16 chips to handle the first two bits of the selector, and a Mux16 chip to handle the remaining three bits. The right pane shows the simulation results. It displays the current values of the inputs and the output. The selector sel is set to 011, which selects input c. The output out is 0001000000000000. The internal pins x and y are also shown, with x being 0001000000000000 and y being 0000000010000000. The bottom bar shows the simulation controls, including a "Test" button and a "Steps: 1 Outputs: 0" indicator.

HDL Code:

```
5 /**
14 *      h if sel = 111
15 */
16 CHIP Mux8Way16 {
17     IN a[16], b[16], c[16], d[16],
18        e[16], f[16], g[16], h[16],
19        sel[3];
20     OUT out[16];
21
22     PARTS:
23     //// Replace this comment with your code.
24     Mux4Way16(a=a, b=b, c=c, d=d, sel=sel[0..1], out=x);
25     Mux4Way16(a=e, b=f, c=g, d=h, sel=sel[0..1], out=y);
26     Mux16(a=x, b=y, sel=sel[2], out=out);
27 }
```

Simulation Results:

Signal	Value (16 bits)	Dec
g	0000000100000000	dec
h	0000000010000000	dec
sel	011	dec
Output pins		
out	0001000000000000	dec
Internal pins		
x	0001000000000000	dec
y	0000000010000000	dec

Test Controls:

Test Script | Compare File | Output File | Diff Table

81
82 set sel 6,
83 eval,
84 output;
85
86 set sel 7,
87 eval,

14. DMux4Way.hdl

```
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/DMux4Way.hdl
/**
 * 4-way demultiplexor:
 * [a, b, c, d] = [in, 0, 0, 0] if sel = 00
 *                  [0, in, 0, 0] if sel = 01
 *                  [0, 0, in, 0] if sel = 10
 *                  [0, 0, 0, in] if sel = 11
 */
CHIP DMux4Way {
    IN in, sel[2];
    OUT a, b, c, d;

    PARTS:
    //// Replace this comment with your code.
    DMux(in=in, sel=sel[1], a=a, b=cd);
    DMux(in=a, sel=sel[0], a=a, b=b);
    DMux(in=cd, sel=sel[0], a=c, b=d);
}
```

Simulator screenshot

The screenshot shows the Nand2Tetris Hardware Simulator interface. The top bar displays the project name "NAND2Tetris / Hardware Simulator / DMux4Way.hdl". The main window is divided into two panes. The left pane shows the HDL code for the DMux4Way chip, with line numbers 5 through 21. The code defines a 4-way demultiplexor that takes an input 'in' and a 2-bit selector 'sel' and outputs four signals 'a', 'b', 'c', and 'd'. The right pane shows the chip's configuration, including input pins 'in' and 'sel', output pins 'a', 'b', 'c', and 'd', and internal pins. The 'in' pin is set to 0, and the 'sel' pin is set to 00. The output pins are also shown with their current values. The bottom status bar indicates the simulation is running at "Steps: 1 Outputs: 0" with a "Slow" speed setting. Below the status bar, there are tabs for "Test Script", "Compare File", "Output File", and "Diff Table". The "Test Script" tab is active, showing a sequence of commands: "35", "36 set sel %B10,", "37 eval,", "38 output;", "39", "40 set sel %B11,", "41 eval,".

15. DMux8Way.hdl

```
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/DMux8Way.hdl
/**
 * 8-way demultiplexor:
 * [a, b, c, d, e, f, g, h] = [in, 0, 0, 0, 0, 0, 0, 0] if sel = 000
 *                               [0, in, 0, 0, 0, 0, 0, 0] if sel = 001
 *                               [0, 0, in, 0, 0, 0, 0, 0] if sel = 010
 *                               [0, 0, 0, in, 0, 0, 0, 0] if sel = 011
 *                               [0, 0, 0, 0, in, 0, 0, 0] if sel = 100
 *                               [0, 0, 0, 0, 0, in, 0, 0] if sel = 101
 *                               [0, 0, 0, 0, 0, 0, in, 0] if sel = 110
 *                               [0, 0, 0, 0, 0, 0, 0, in] if sel = 111
 */
CHIP DMux8Way {
    IN in, sel[3];
    OUT a, b, c, d, e, f, g, h;

    PARTS:
    //// Replace this comment with your code.
    DMux(in=in, sel=sel[2], a=x, b=y);
    DMux4Way(in=x, sel=sel[0..1], a=a, b=b, c=c, d=d);
    DMux4Way(in=y, sel=sel[0..1], a=e, b=f, c=g, d=h);
}
```

Simulator screenshot

The screenshot shows the Nand2Tetris Hardware Simulator interface. The top bar displays "NAND2Tetris / Hardware Simulator / DMux8Way.hdl". Below the bar, the HDL code for DMux8Way.hdl is visible in the editor. The code defines an 8-way demultiplexor chip with an input 'in', a 3-bit selector 'sel', and eight outputs 'a' through 'h'. The code uses a series of multiplexers to route the input to the correct output based on the selector value.

On the right side of the simulator, the "Output pins" section shows the current values of the outputs. The inputs are set to 'in = 1' and 'sel = 0 1 1' (binary 110). The output pins are labeled 'a' through 'g' (with 'h' not shown), and their values are: a=0, b=0, c=0, d=1, e=0, f=0, g=0. This indicates that the input 'in' is being routed to output 'd'.

At the bottom of the simulator, there is a "Test" section with a "Test Script" tab. The script shows the following code:

```
67
68 set sel %B110,
69 eval,
70 output;
71
72 set sel %B111,
73 eval,
```